

Speaker 3 of 5 — Data Cleaning

Career Compass SG · Module 1 Assignment · Group 7

Your slot: 3:45 – 6:00 (2 minutes 15 seconds) · You have the strongest material

Your job in one sentence

Prove that we understood this data rather than just loading it. You own the single best story in the whole presentation — the engagement break — and the assessment gives 25% to data handling, the largest single weighting. Spend your time on the *reasoning*, not the inventory of rules.

Where you sit in the flow

	Speaker	Topic
	1	Dashboard overview
	2	Technology stack
→ YOU	3	Data cleaning
	4	Data analysis 1 — market & pay
	5	Data analysis 2 — competition, skills & scoring

Speaker 2 hands you: "...none of it matters if the numbers going in are wrong — and this dataset had a trap in it." **You hand to Speaker 4** with: "So that's the data we trust. Here's what it actually says."

Discipline rule: you are the most likely person to overrun. **If the clock passes 5:00 and you haven't started Part C, skip straight to it.** Part C is the part that must land.

Script

Part A — The principle, not the inventory (3:45 – 4:25)

(Slide: cleaning decisions — you will walk 3 rows, not 11)

Eleven cleaning rules. I'll give you the **principle** instead: **blank the untrustworthy field, keep the trustworthy row**. A posting with a broken salary still proves a job exists.

- **occupationId** was 100% empty across all 1.05 million rows — dropped.
- **10,016 postings claimed a monthly salary above \$60,000 or below \$500** — annual figures in a monthly field. We blanked the salary and kept the posting.
- **We capped the top and bottom 1%** rather than deleting, because deleting would have biased our demand counts.

99.6% of rows survive.

Part B — Two traps worth naming (4:25 – 4:55)

(Slide: the duplicates trap + the two-table diagram)

Two traps.

.duplicated() lied to us — it reported 289 duplicate job IDs; all 289 were repeated **blanks**. There are **zero real duplicates**.

And one job belongs to up to three categories — 1.69 on average — so counting rows and counting category-rows answer different questions. We shipped that bug to ourselves: for an hour our overview said **1.66 million postings instead of 1.04**.

Part C — The finding that changed the product (4:55 – 6:00)

(Slide: the two-line engagement chart — this is your centrepiece, do not rush it)

Now the finding that changed everything.

We plotted views and applications over time as a sanity check. Look at **June 2023**: mean views per posting falls **from about 111 to about 5**. Zero-application postings jump **from 18% to 79%**.

That's not Singapore losing interest. **This dataset is two extracts stitched together** — everything after July 2023 was captured at posting time, so those counters never accumulated.

So demand and salary use all 1.04 million rows, but every competition metric uses only postings up to 30 June 2023 — 203,702 of them — and it says so on screen every time.

If we'd missed this, every track would have looked uncontested, and our recommender would have pointed people at the most crowded markets in Singapore.

Same story, smaller: **volume ramps until May 2023**, so every trend figure starts there.

That's the data we trust. Here's what it says.

Numbers you must know cold

Figure	Value
Rows read → rows kept	1,048,585 → 1,044,597 (99.6%)
Wholly empty rows dropped	3,988 (0.38%)
Impossible salaries blanked	10,016
Rows capped at 1st/99th percentile	19,647 (band \$1,150 – \$16,500)
Genuine duplicate job IDs	0 (the 289 were repeated blanks)
Categories per posting	1.69 average , max 3 → 1,767,829 long rows
Engagement break	views 111 → 5; zero-application 18% → 79%, at Jun/Jul 2023
Competition window	postings ≤ 2023-06-30 , 203,702 of them
Volume reliable from	May 2023 (~75,000 postings/month)
Salary disclosed after cleaning	99.0%

The eleven rules, if anyone asks for the full list

Issue	Decision
<code>occupationId</code> 100% null	drop column
<code>status_id</code> duplicates <code>status_jobStatus</code>	drop column
3,988 wholly empty rows	drop rows
<code>salary == 0</code>	set to missing — zero means "not disclosed"
<code>salary < \$500</code> or <code>> \$60,000/month</code>	blank the salary, keep the row
<code>salary_min > salary_max</code>	swap the values — fields entered backwards
long right tail	cap at 1st/99th percentile
<code>experience > 40</code> years	set to missing
<code>vacancies 0</code> or <code>> 500</code>	floor at 1 / blank above 500
duplicate job IDs	de-duplicate <code>keep="last"</code> (no-op here)
<code>categories</code> JSON array	parse and explode to a long table

Questions you own

"Why cap outliers instead of dropping them?"

Dropping the row removes the posting from the demand count too. Capping keeps the demand signal honest while stopping 19,647 extreme values from steering every average.

"How do you know the June 2023 break is a collection artefact and not real?"

Three things move together at exactly one boundary: mean views, zero-application share, and nothing else — posting volume, salaries and categories are all continuous across it. A real collapse in jobseeker interest would show up in the postings too. It doesn't.

"Isn't throwing away 80% of your data for competition metrics a problem?"

We're not throwing it away — demand and salary still use all of it. For competition we use 203,702 postings, which is an ample base. The alternative was using numbers we know are wrong.

"Why `repost_count >= 1` for hard-to-fill and not `>= 2`?"

Because the field is capped at 2 in this extract — the only values are 0, 1 and 2. At `>= 2` we'd flag 1.4% of postings; at `>= 1`, 4.1%. Any repost at all is the meaningful signal here.

"What about the 0.14% of missing values — why drop rows rather than impute?"

We checked first whether the gaps were the *same* rows. They were: wholly empty export lines with no ID, no title, no company and no dates. There's nothing to impute from.

Rehearsal checklist

- [] Timed at **2:15** — and you know which part to cut if you're over (Part A drops to one example)
- [] You can deliver the engagement-break story **without the slide**, from memory
- [] You can say why it's an artefact, not a real signal, in two sentences
- [] You are comfortable saying "we shipped that bug to ourselves" — owning a mistake reads as competence, not weakness
- [] Handover line to Speaker 4 practised